LESSON 3.2.3

Data Safety Measures

This lesson covers how AI systems are exposed to data-related risks and the protective measures used to mitigate them. Topics include data breaches, model bias, masking, anonymization, redaction, minimization, data loss prevention, and regulatory compliance.

Risk and AI Systems

Risk is the potential for threats and vulnerabilities to compromise an information system's **confidentiality**, **integrity**, and **availability** (CIA).

Confidentiality

Ensuring data is accessible only to authorized users

Integrity

Ensuring data remains accurate and unaltered

Availability

Ensuring systems and data are accessible when needed

AI systems are not immune — risks emerge from how they **access**, **utilize**, and **store** data throughout their lifecycle.

Data Breach

A **data breach** occurs when data is accessed or exposed to unauthorized users. Breaches can originate from inside or outside an organization.

Lack of Encryption

Data exposed during storage or transfer because encryption was not applied

Internal Threat

An employee with incorrect access permissions exposes sensitive data

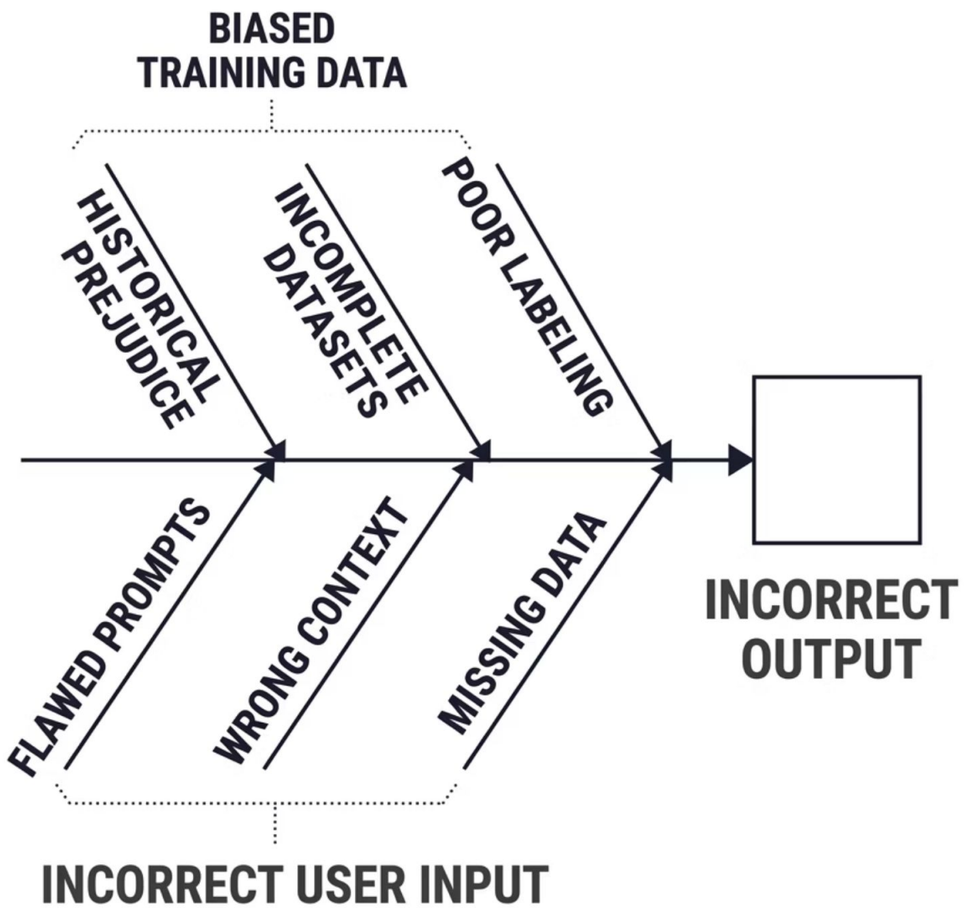
External Threat

A malicious actor gains unauthorized access from outside the organization

Model Bias

Model bias is a critical risk in AI systems. When training data is biased, incomplete, or incorrect or when users provide flawed inputs, the model produces unreliable outputs.

> This can lead to **unfair outcomes**, **discriminatory decisions**, or **costly business errors**. The guiding principle is simple:
>
> **Garbage in, garbage out.**



Data Masking

Data masking replaces **real information** with **fictional but realistic values**. It is used when actual data content is unnecessary such as in testing, development, or non-production environments.

Static Masking

Creates a **separate dataset** where masking values are **permanently applied**. The original data is never exposed in that copy.

Dynamic Masking

Masks data **in real time** as it is accessed or used, then **returns to its original value** once at rest. The source data is never permanently altered.

Anonymization, Redaction, and Minimization

Three related but distinct techniques for limiting data exposure:

Anonymization

Removes **identifying details** from a dataset. For example, removing birth day and year when only birth month is required.

Redaction

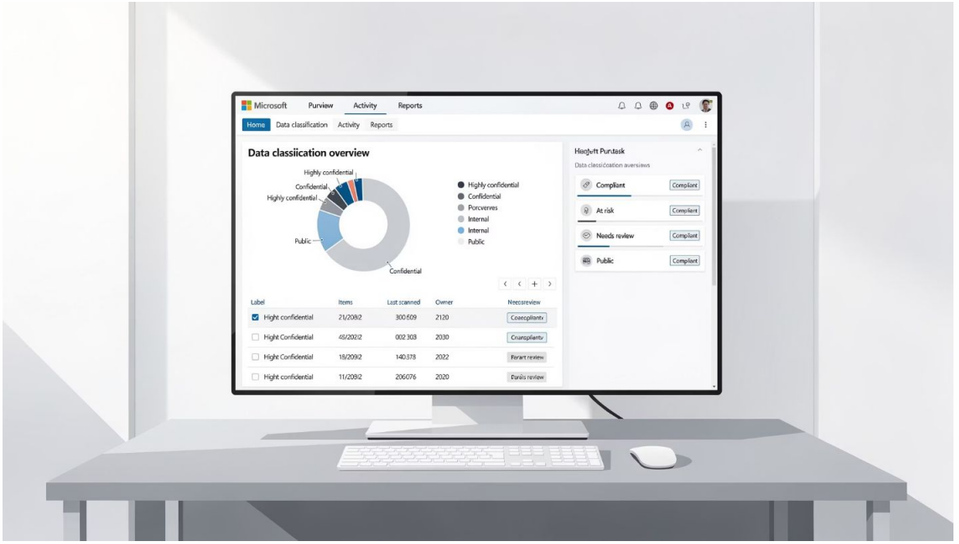
Removes, obscures, or masks data that is **no longer necessary after collection**. Applied post-collection.

Minimization

Collects only the **bare minimum data** needed for a task. Applied **before or during** collection not after.

Key Difference: Minimization limits what is collected upfront; redaction removes unnecessary data after the fact.

Data Loss Prevention (DLP) and Classification



Public

Internal

Confidential

Secret

Top Secret

DLP programs such as **Microsoft Purview DLP** or **Trellix Data Loss Prevention** monitor data across its entire lifecycle, from collection to destruction.

They require data to be **classified and labeled** to function effectively. Common classification labels include:

Regulatory Compliance

Compliance frameworks such as **GDPR**, **HIPAA**, and **CCPA** establish requirements for the safe collection, use, transport, and storage of data. If an AI system processes regulated data, these controls **must be in place**.

⚠️ Non-Compliance Risk

Can result in **financial fines** or **restrictions** on data collection activities

🛡️ Best Practice

Minimize the **amount and type** of data retained to reduce exposure in the event of a breach

✅ Required Controls

Encryption, access controls, audit logs, and data classification must all align with applicable regulations

Key Takeaways

Review these core concepts from Lesson 3.2.3 before moving on.

1

AI Risk & CIA Triad

AI systems face risks to confidentiality, integrity, and availability of data

2

Data Breaches

Can originate internally (improper access) or externally (malicious actors); encryption is a key defense

3

Model Bias

Biased or flawed training data leads to incorrect, unfair, or harmful AI outputs

4

Data Masking

Static masking is permanent; dynamic masking is temporary and reverts once data is at rest

5

Anonymization, Redaction & Minimization

Three post- and pre-collection techniques that reduce data exposure and limit identifying information

6

DLP & Classification

DLP tools monitor the full data lifecycle; classification labels (Public → Top Secret) enable enforcement

7

Regulatory Compliance

Non-compliance can result in fines or restrictions; minimize retained data to reduce breach risk